

EDUCATION

University of Massachusetts Dartmouth — B.S. Computer Science · GPA 3.5 Expected May 2027 · North Dartmouth, MA

TECHNICAL SKILLS

GPU / Systems	CUDA, Triton, PyTorch internals, kernel profiling (Nsight Compute, Nsight Systems), memory coalescing & tiling, quantization (INT8 PTQ), FlashAttention / attention kernel implementation
Languages	Python, C/C++, SQL
ML & Deep Learning	PyTorch, TensorFlow/Keras, Transformers & attention mechanisms, CNNs, object detection (YOLO, RT-DETR), anomaly detection (PatchCore), transfer learning
Cloud & Backend	AWS (EC2, Lambda, S3, API Gateway), Snowflake, FastAPI, Docker

EXPERIENCE

Fleet Robotics — ML Engineer Intern Summer 2026 · Boston, MA

- Diagnosed a sim-to-real heading-drift discrepancy in a MuJoCo robotics RL policy: measured 1.3°/pivot drift compounding to ~20° over 15 steps; corrected the simulator's drift model (16.6° predicted), validated against 20 real robot runs
- Retrained a SAC reinforcement learning policy building on an existing production policy
- Built data engineering pipelines (AWS, Snowflake) and Streamlit dashboards turning raw robot telemetry into engineering insights
- Navigated and contributed to a ~500K-line production codebase; presented findings to engineering and non-technical stakeholders

SMAST – Gavin Fay Lab — Research Software Engineering Intern Sept – Nov 2025 · Dartmouth, MA

- Refactored R codebase for NOAA-funded SSEEP project evaluating offshore wind impacts on fishery surveys
- Implemented parallel processing techniques, achieving 2x speedup in computation time for spatial simulations
- Developed and packaged sseeph helpers — an R library to streamline common data operations

SMAST — Data Science Intern May – Aug 2025

- Automated analysis of 38 years of estuary data (10K+ records) across 7 sites, building scalable workflows that cut manual effort by 70%
- Developed PowerQuery ETL pipelines, reducing report turnaround from 3 weeks to 10 minutes (95% faster)

Blue Tech Externship (Courage Builder, 2025 & 2026) — built CMVS, a simulated autonomous UUV for sonar-based mine detection; 1st place, 2025 cohort

PERSONAL PROJECTS

cpp-gpu-inference — GPU Inference Stack Built From Scratch CUDA · Triton · PyTorch · C++

- Achieved ~95% of FP16 peak throughput at sequence length 8192 with a FlashAttention-2 implementation (forward + backward) written from scratch in Triton, wrapped in a custom torch.autograd.Function and integrated into a from-scratch transformer
- Built INT8 post-training quantization from scratch (vector-wise quantization + mixed-precision decomposition); benchmarked vs. bitsandbytes on a 200K-pair translation model — 15/15 identical outputs, perplexity within 0.02%, 46.4% static memory reduction (vs. 29.3%)
- Wrote and autotuned custom Triton matmul kernels, benchmarked within 20% of cuBLAS across problem sizes 512–4096, exceeding cuBLAS at smaller sizes
- Built CUDA kernels from first principles (PMPP): memory coalescing, shared-memory tiling, thread coarsening; profiled with Nsight Compute/Systems

XrayShield AI — Screener Performance Intelligence Platform PyTorch · PatchCore · React · FastAPI

- Co-founded a startup building a screener performance intelligence platform for airport X-ray security; trained a PatchCore anomaly detection model on 100GB+ of baggage imagery across 6 public datasets, generating pixel-level threat heatmaps
- Built a React screening interface with real-time AI heatmap overlay, decision-timing analytics, and session-level fatigue tracking; backend with FastAPI, SQLite, and Docker